

Student Name: _____

FURTHER MATHEMATICS

Unit 3

Targeted Evaluation Task for School-assessed Coursework 1



2020 Application Task on Data Analysis for Outcomes 1, 2 & 3

Recommended writing time*: 300 minutes

Total number of marks available: 40 marks

TASK BOOK

* The recommended writing time is a guide to the time students should take to complete this task. Teachers may wish to alter this time and can do so at their own discretion.

Conditions and restrictions

- Students are permitted to bring into the room for this task: pens, pencils, highlighters, erasers, sharpeners and rulers, a CAS calculator and/or scientific calculator.
- Students are NOT permitted to bring into the room for this task: blank sheets of paper and/or white out liquid/tape.

Materials supplied

- Question and answer book of 11 pages.

Instructions

- Print your name in the space provided on the top of the front page.
- All written responses must be in English.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic communication devices into the room for this task.

Australia is one of five countries who have competed in every Summer Olympic Games (competing with New Zealand as Australasia in 1908 and 1912) and despite the relatively small population, sits eighth on the Summer Olympic Games medal tally with 513 medals.

The number of events, and subsequently number of medals that are able to be won, have increased over the years. The first Modern Olympic Games in 1896 had only 43 events whereas the 2020 Olympics in Tokyo will feature 339 events including five new sports and fifteen new events that will be held for the first time.

Australia's interest in the Olympic Games increased when we hosted our first Olympic Games in Melbourne in 1956. The Melbourne Olympics were arguably Australia's most successful, finishing 3rd on the medal tally, winning a total of 35 medals, which was only surpassed 40 years later in Atlanta 1996. In Melbourne in 1956 Australia won 13 gold medals, which was not surpassed until we hosted our second Olympic Games in Sydney 2000 when 16 gold medals were won.

Australia's success in the Olympic Games continued in the 1960, 1964, 1968 and 1972 Olympics, where we remained in the top ten on the final medal tally. The 1976 Montreal Olympics were one of Australia's least successful games, winning one silver and four bronze medals. In response to Australia's performance in these Olympics, the Australian Government created the Australian Institute of Sport, which saw a steady increase in the number of medals, in particular gold medals, that Australia has won in the following Olympic Games.

The following investigation looks at Australia's performance in the Olympic Games.

Randomly select a series of 15 consecutive Olympiads to use for your data in this investigation.

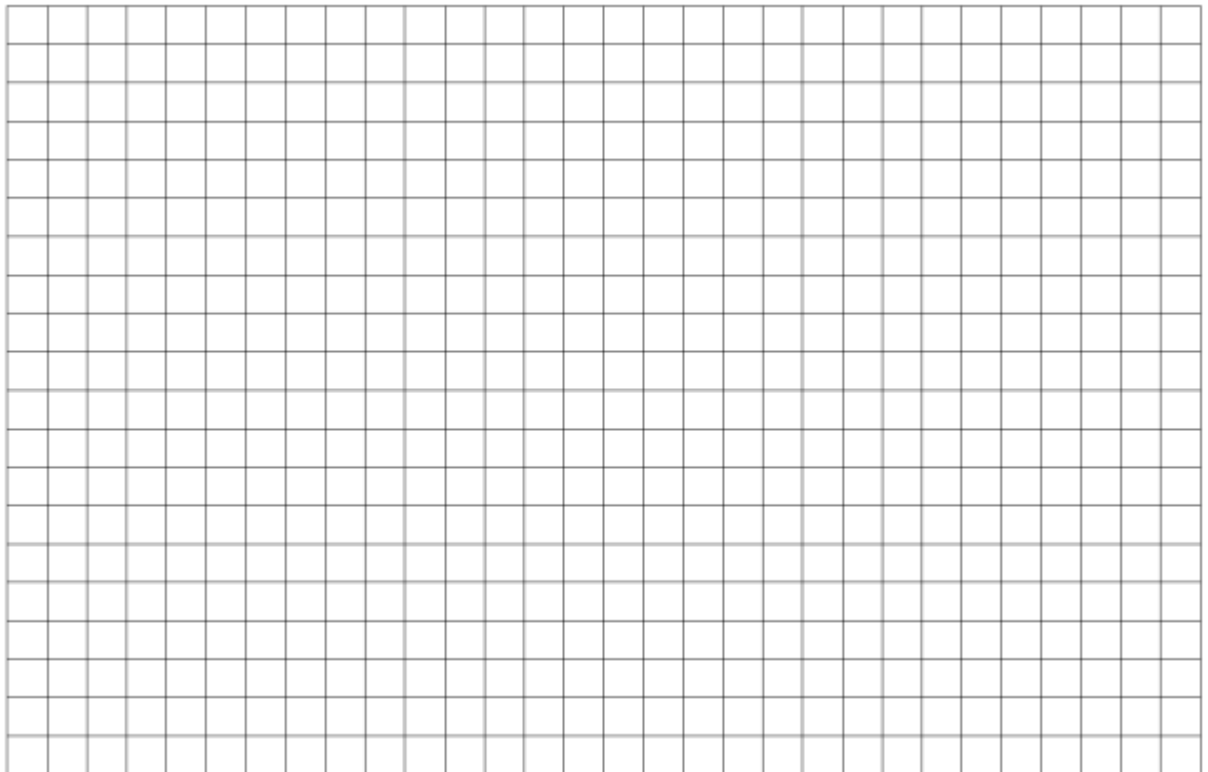
Starting Olympiad (from 1-13) _____

2020 FURTHER MATHEMATICS SCHOOL-ASSESSED COURSEWORK

Olympiad	Olympic Games	Athletes	Gold	Silver	Bronze	Total	Rank
1	 1896 Athens	1	2	0	0	2	8
2	 1900 Paris	3	2	0	3	5	9
3	 1904 St. Louis	3	0	3	1	4	10
4	 1908 London	32	1	2	2	5	11
5	 1912 Stockholm	26	2	2	3	7	12
6	 1920 Antwerp	13	0	2	1	3	16
7	 1924 Paris	37	3	1	2	6	11
8	 1928 Amsterdam	18	1	2	1	4	19
9	 1932 Los Angeles	12	3	1	1	5	10
10	 1936 Berlin	33	0	0	1	1	30
11	 1948 London	77	2	6	5	13	14
12	 1952 Helsinki	85	6	2	3	11	9
13	 1956 Melbourne	314	13	8	14	35	3
14	 1960 Rome	188	8	8	6	22	5
15	 1964 Tokyo	234	6	2	10	18	8
16	 1968 Mexico City	175	5	7	5	17	9
17	 1972 Munich	173	8	7	2	17	6
18	 1976 Montreal	184	0	1	4	5	32
19	 1980 Moscow	123	2	2	5	9	15
20	 1984 Los Angeles	240	4	8	12	24	14
21	 1988 Seoul	270	3	6	5	14	15
22	 1992 Barcelona	290	7	9	11	27	10
23	 1996 Atlanta	424	9	9	23	41	7
24	 2000 Sydney	630	16	25	17	58	4
25	 2004 Athens	482	17	16	17	50	4
26	 2008 Beijing	433	14	15	17	46	6
27	 2012 London	410	8	15	12	35	8

Question 1 (10 marks)

- a. Using appropriate univariate analysis techniques, summarise and describe the distribution of the number of gold medals Australia won over your chosen 15 Olympiad period. The following aspects must be addressed and described; mean, median, standard deviation, distribution, a suitable graphical representation.



4 marks

- b.** Construct an appropriate display to describe the trend of the number of gold medals won over the 15 Olympiads (Use the Olympiad numbers not the years, as the 1916, 1940 and 1944 Olympic Games were not held due to war.).

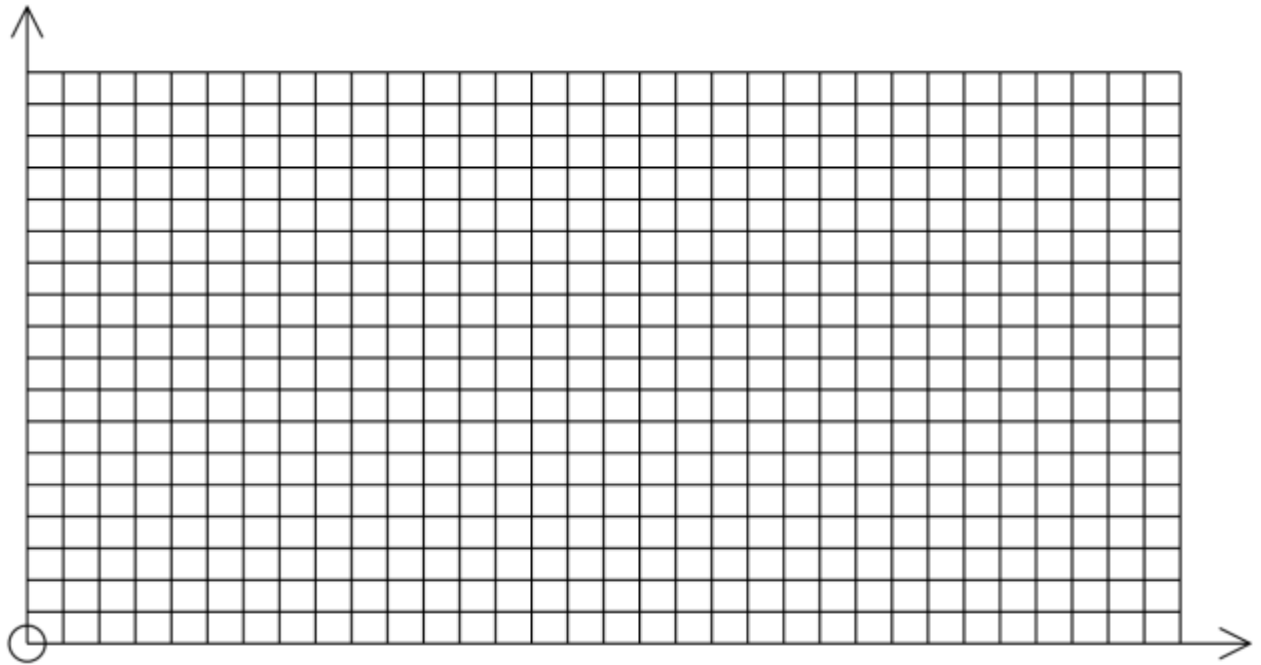
3 marks

- c.** Discuss the relationship shown in the display in part **b.**

3 marks

Question 2 (18 marks)

- a. Determine the equation of the regression line for the **total number of medals won** for Olympiads one to twenty-seven (Athens 1896 to London 2012), interpret the summary regression statistics in context, and draw the regression line on the scatterplot. Including the interpretation of the regression coefficients in the terms of the variables and commenting on any outliers in your data. Write the regression coefficients correct to four decimal places.



8 marks

- b.** With reference to the shape of the residual plot, comment on whether a linear model is most suitable.

3 marks

- c.** Apply the most suitable transformation to either of the variables (either number of medals or Olympiad), then write the equation of the new regression line and describe the relationship using the new variables. Give your answer using figures correct to four decimal places.

4 marks

- d.** Use your preferred model to predict the total number of medals won in the 2012 London Olympics and 2016 Rio de Janeiro Olympics. Justify why you have used this model and comment on the reliability of your predictions.

The number of medals won in Rio was 29 in total (8 gold, 11 silver and 10 bronze)

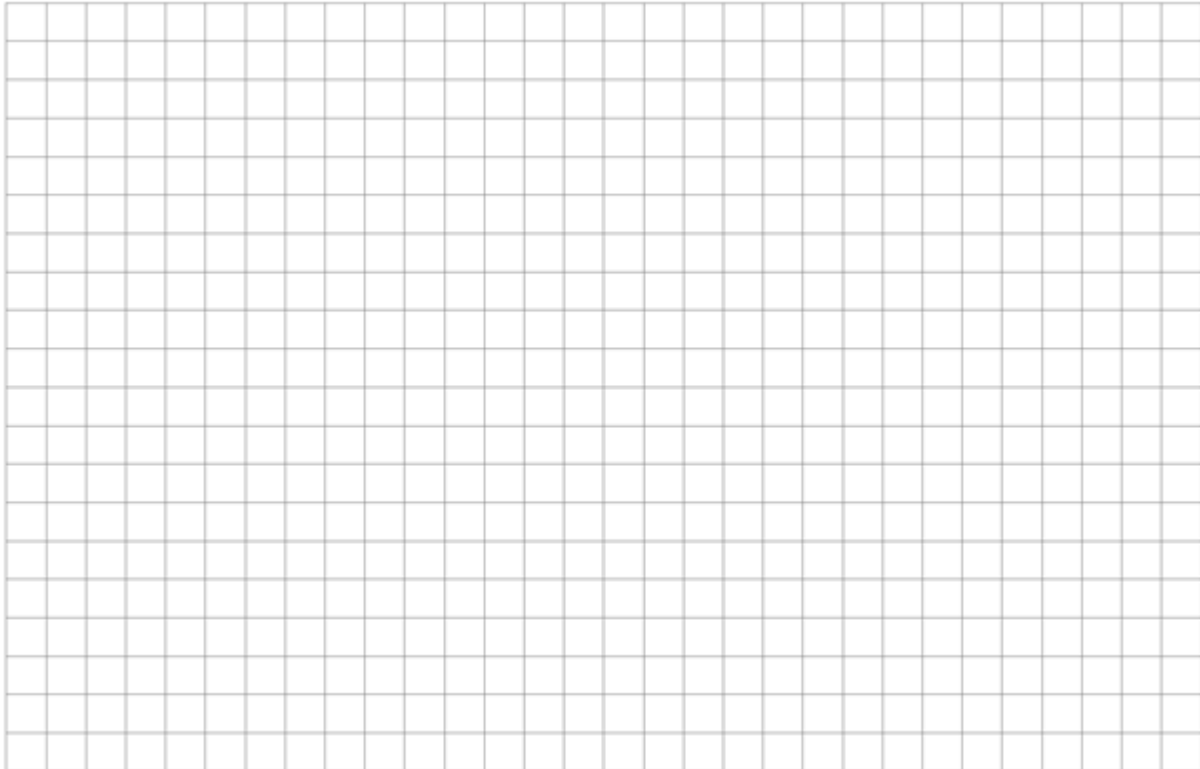
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3 marks

Question 3 (12 marks)

It is suggested that the Olympiad may not be the most reliable predictor of the number of medals that Australia will win at a given Olympics and that the number of participants Australia sends is a better predictor of the total number of medals.

- a. Using CAS, construct a model for the number of medals won against the number of Australians competing in that Olympics and draw the results on the scatterplot below using an appropriate scale for your axis, including the least squares regression line.



4 marks

- b. Calculate the proportion of variation in the number of medals Australia won that can be explained by the number of Australians competing and compare it to the model using your preferred model for the Olympiad.

2 marks

- c. Calculate the predicted number of medals won in the 2012 London Olympics and the 2016 Rio de Janeiro Olympics, based on the number of participants Australia sent to that Olympics. (422 Australian athletes competed in Rio.)

2 mark

- d. Compare the predictions to the real data.

2 mark

- e. Compare this prediction with the prediction based on the Olympiad number and suggest which model would be most accurate in predicting the number of medals Australia will win in the 2020 Olympics

2 mark

END OF TASK BOOK